

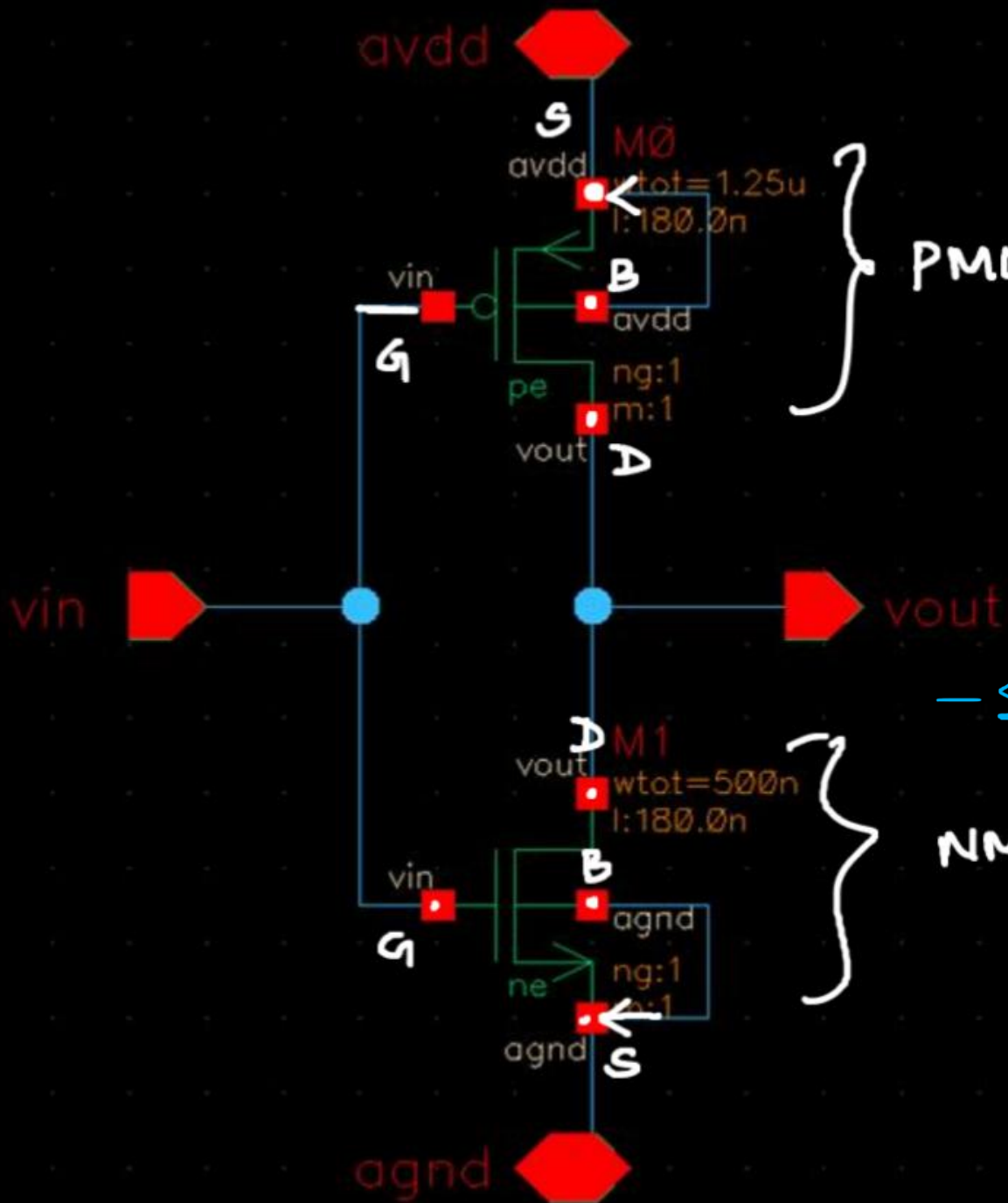
BODY EFFECT & REGIONS OF OPERATION

LECTURE – 3 :: (27-03-2025)

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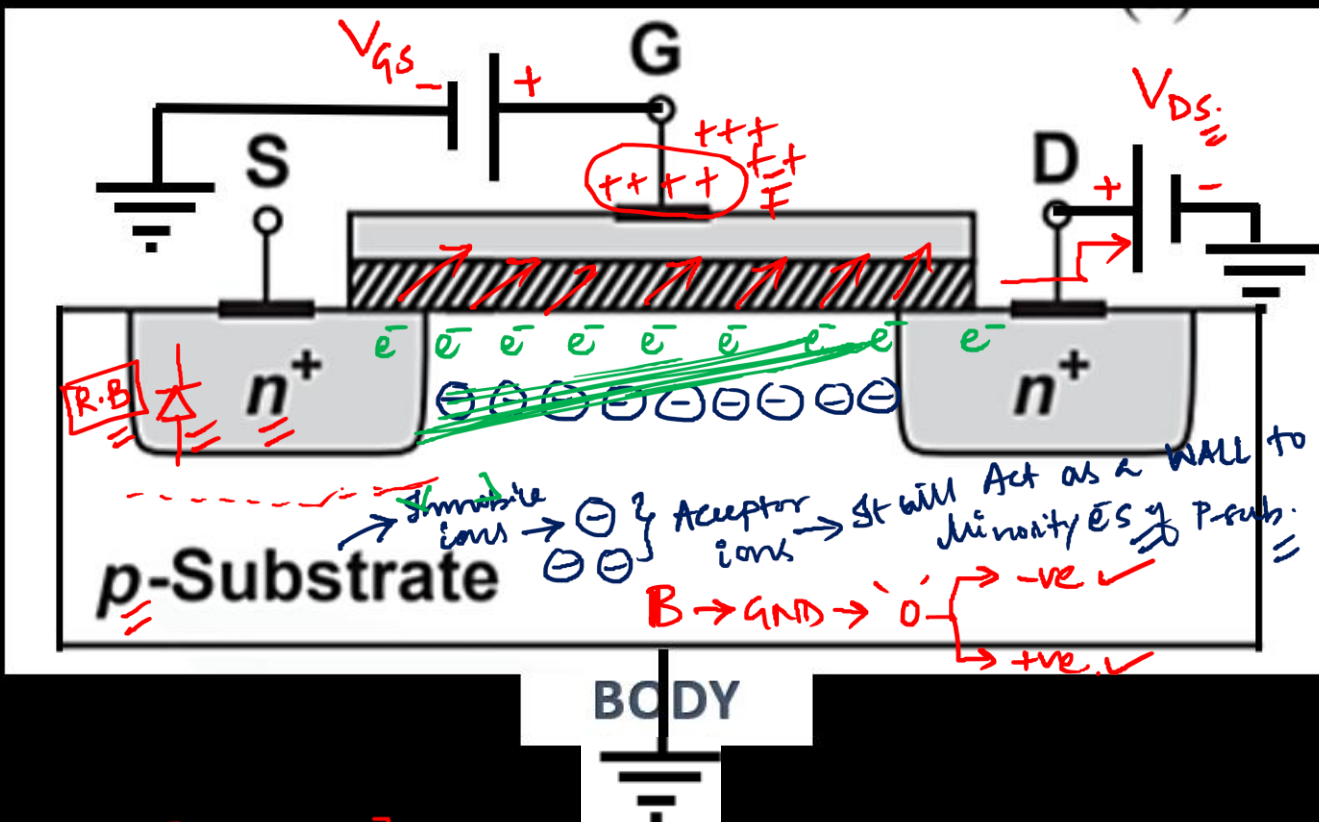
- In PMOS $\rightarrow V_{tp} < 0$ Means Negative 'or' less voltage to turn ON.

PMOS (BODY $\rightarrow$ VDD).
- to prevent BODY EFFECT.

- In NMOS, $V_{tn} \rightarrow +ve \rightarrow$ More potential is required to turn ON the DEVICE !!

NMOS. (BODY $\rightarrow$ GND).
- to prevent BODY EFFECT.

Concept of BODY EFFECT



FERMI-POTENTIAL.

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|} \right)$$

THRESHOLD VOLTAGE @ BULK/BODY = 0V.

BODY BIAS effect Coefficient = $\frac{\sqrt{2 \cdot \epsilon_{Si} \cdot N_A}}{C_{ox}}$

3.9760

CASE-01 $\rightarrow$ if $V_{SB} \uparrow \rightarrow V_{TH} \uparrow = V_{TH0} + \gamma (\neq 0)$

CASE-02 $\rightarrow$ if $V_{SB} \downarrow \rightarrow V_{TH} \downarrow = V_{TH0} - (\neq 0)$

$$\phi_F = \frac{KT}{q} \ln\left(\frac{N_A}{n_i}\right)$$

potential e's $\uparrow$

Body should be grounded / Vdd

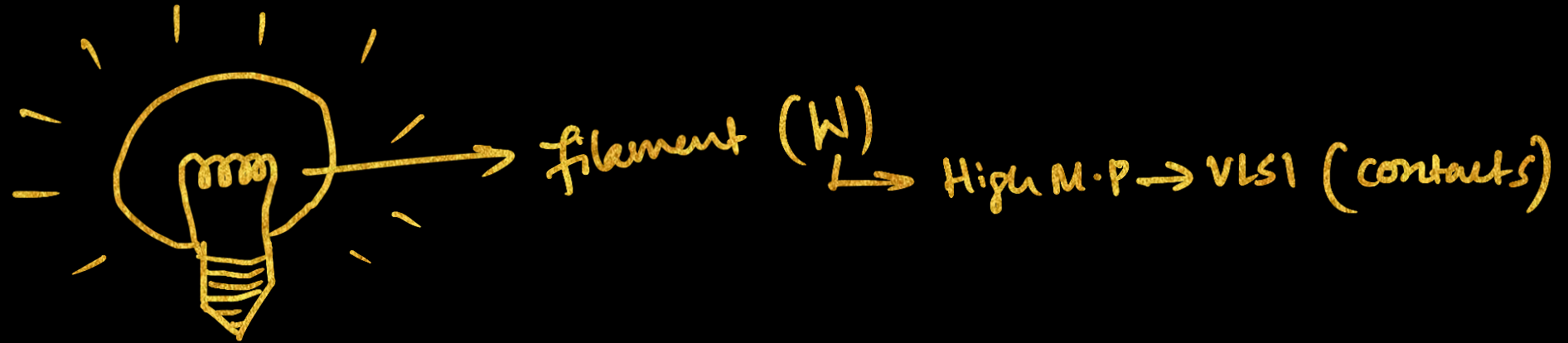
CASE-01 $\rightarrow V_{TH} \uparrow \rightarrow P_{consum} \uparrow \rightarrow$ if impedance $\uparrow$ } CASE-01

$\rightarrow$ VTC will be Abnormal

Typically $[0.5 - 0.6V]$

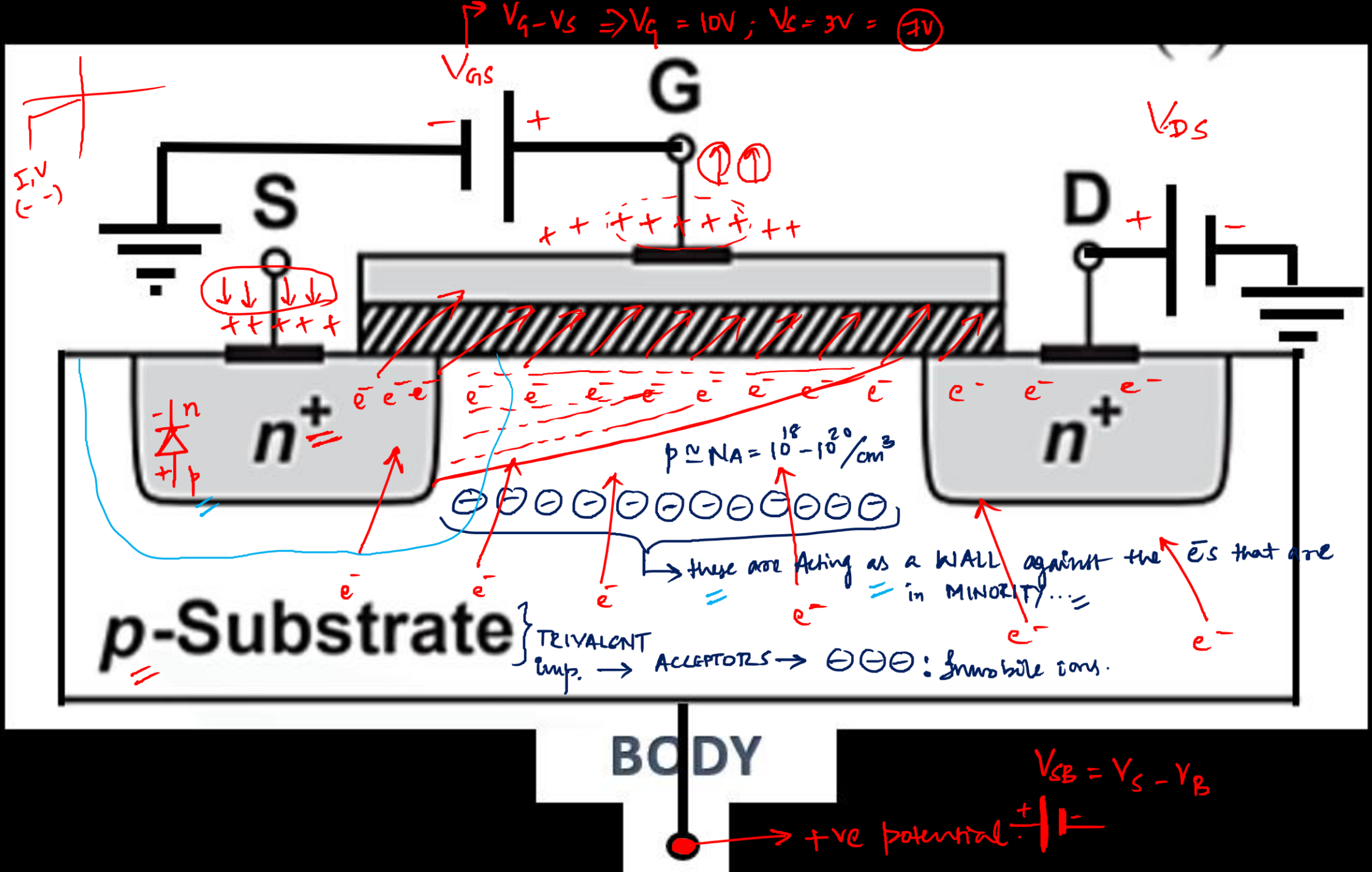
THRESHOLD VOLTAGE

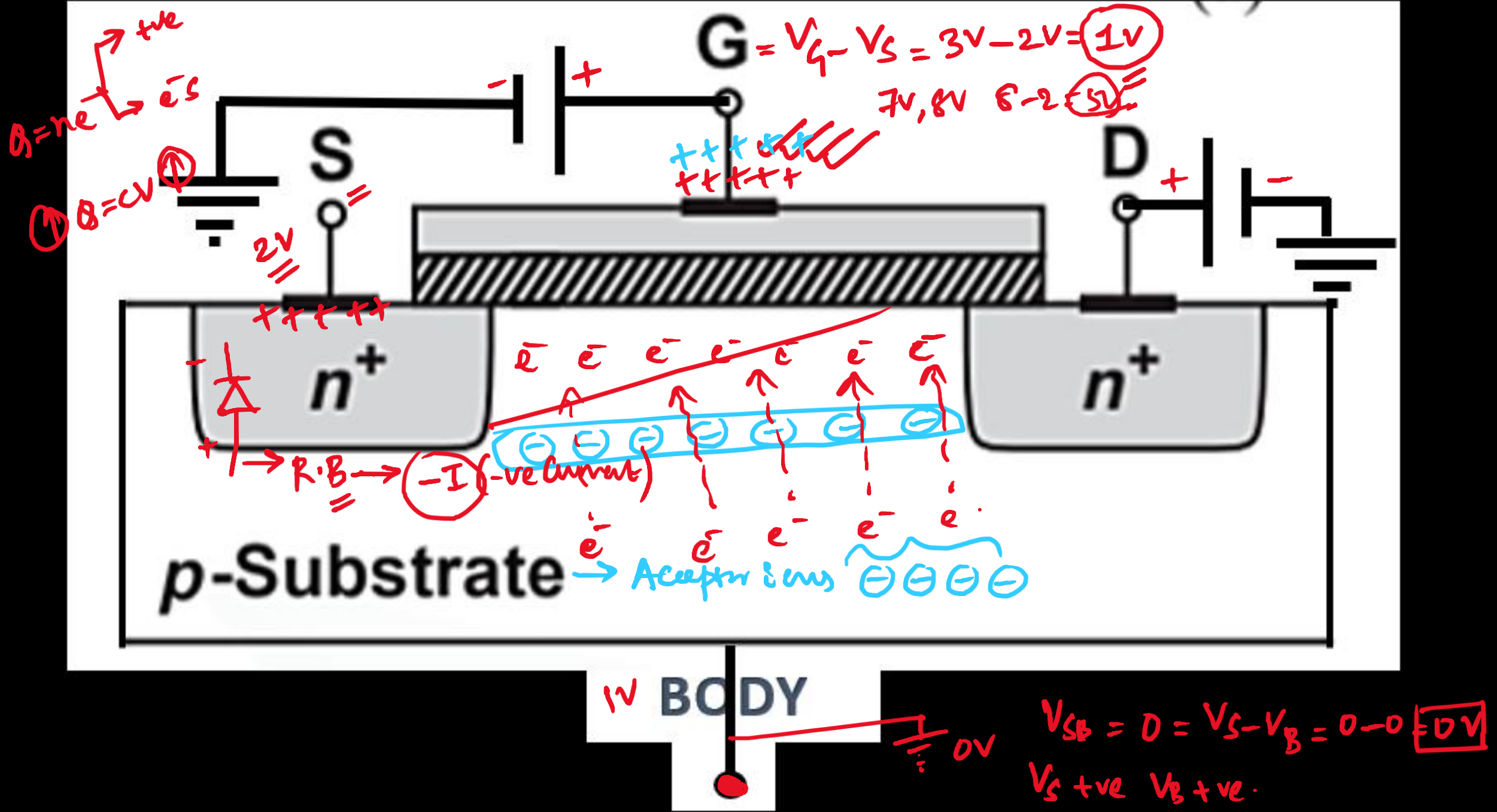
$$V_{TH} = V_{TH0} + \gamma (\sqrt{|2\phi_F + V_{SB}|} - \sqrt{|2\phi_F|}),$$

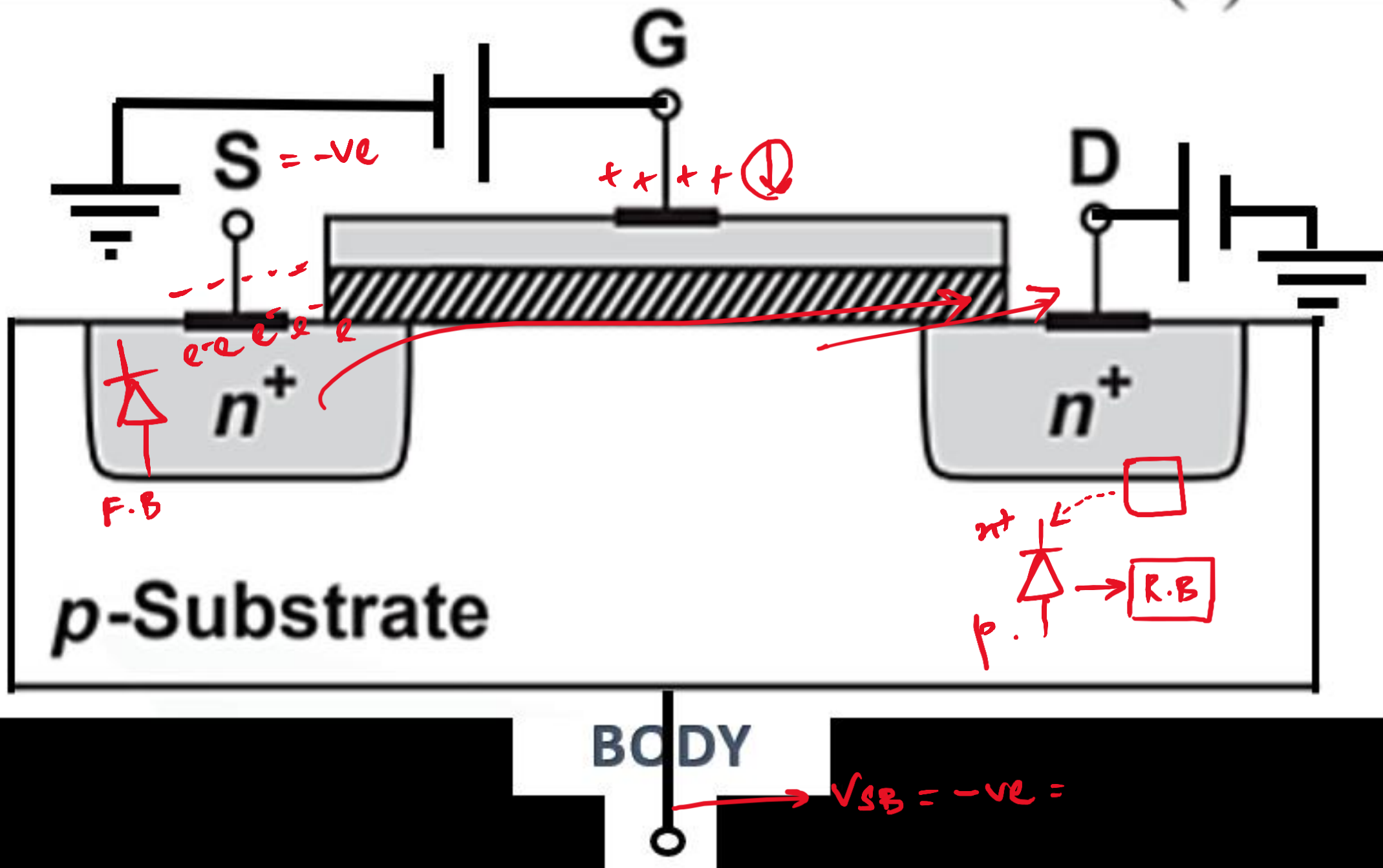


BULB → RATED
POWER → $V \cdot I = 200 \text{ V} - 1.5 \text{ A} = \boxed{3000 \text{ W.}}$

① ↓ ✓
frequency = ... KHz
MHz







CASE-02 :-

$$V_{TH} = V_{TH0} + \gamma \left(\sqrt{2\phi_F + V_{SB}} \right) + \sqrt{2\phi_F}$$

- (i) $V_{SB} \uparrow \Rightarrow V_{TH} \uparrow$; Less potential will turn ON the device
- (ii) Noise immunity will be less / More prone to NOISE.

Why (1) PMOS $\rightarrow$ Bulk $\rightarrow$ V_{dd} $\therefore = 1.8V \rightarrow V_{tp} < 0$
(2) NMOS $\rightarrow$ Bulk $\rightarrow$ V_{ss} $\therefore = 0V \rightarrow V_{tn} > 0$ } just remind me !!

OPERATING REGIONS of MOSFETS

NMOS

- | | |
|----|----------------|
| 1. | ACCUMULATION ✓ |
| 2. | DEPLETION ✓ |
| 3. | INVERSION ✓ |

↓
TO TURN 'ON'
the DEVICE

- | | |
|----|------------------------|
| 1. | CUT OFF → Switch |
| 2. | LINEAR → Resistor / SW |
| 3. | SATURATION → (↑) |

↓
BIASING
=

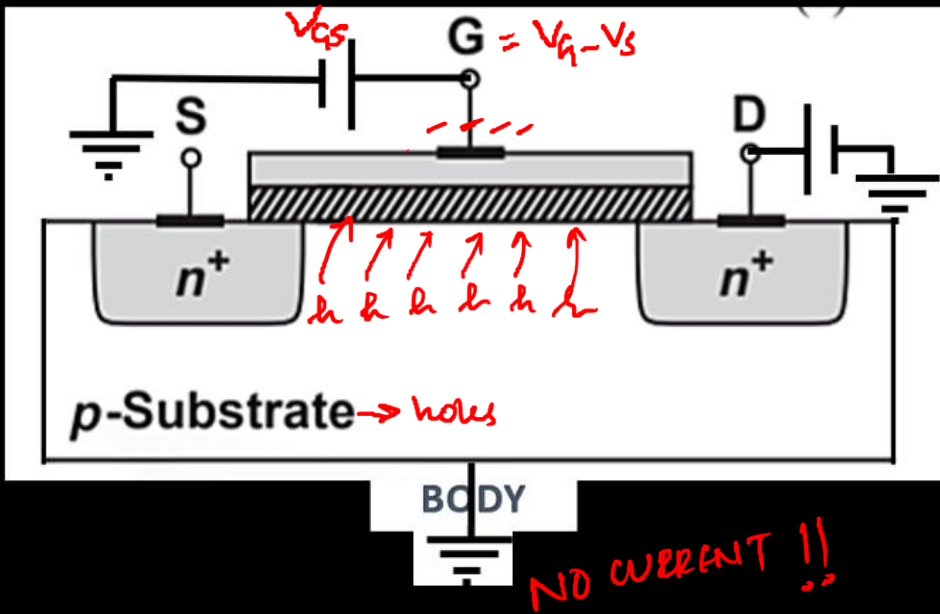
→ Amplifier ✓

+ In sub-threshold region =

PMOS

- | | |
|----|--------------|
| 1. | ACCUMULATION |
| 2. | DEPLETION |
| 3. | INVERSION |

- | | |
|----|------------|
| 1. | CUT OFF |
| 2. | LINEAR |
| 3. | SATURATION |



ACCUMULATION

$$V_{GS} - V_{DS} < 0 \quad \boxed{OA} =$$

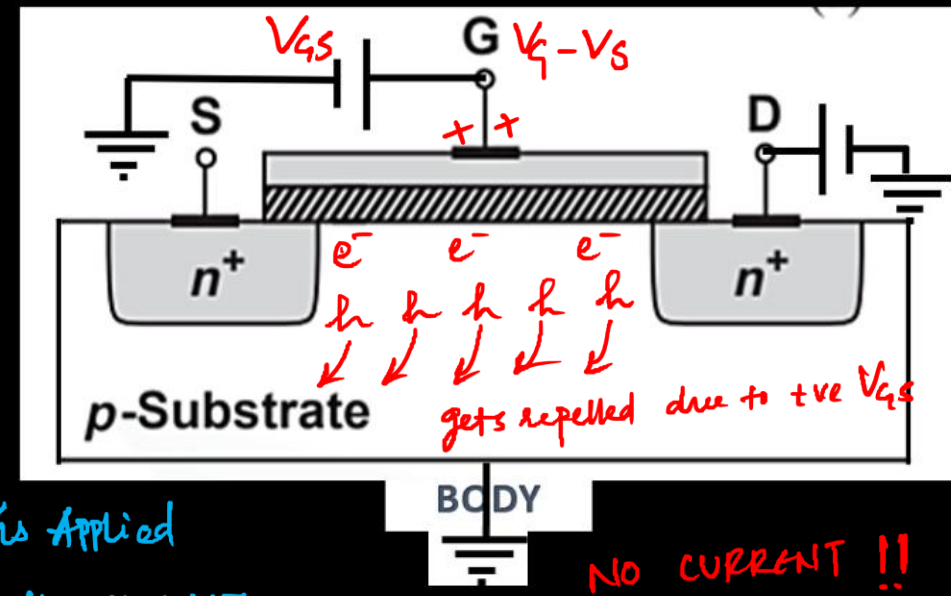
→ The Accumulation of Majority charge carriers i.e. Holes get ACCUMULATED just under the Gate electrode inside the device

This phenomenon is known as

ACCUMULATION; Here; There is

NO CHANNEL FORMATION

NMOS



DEPLETION

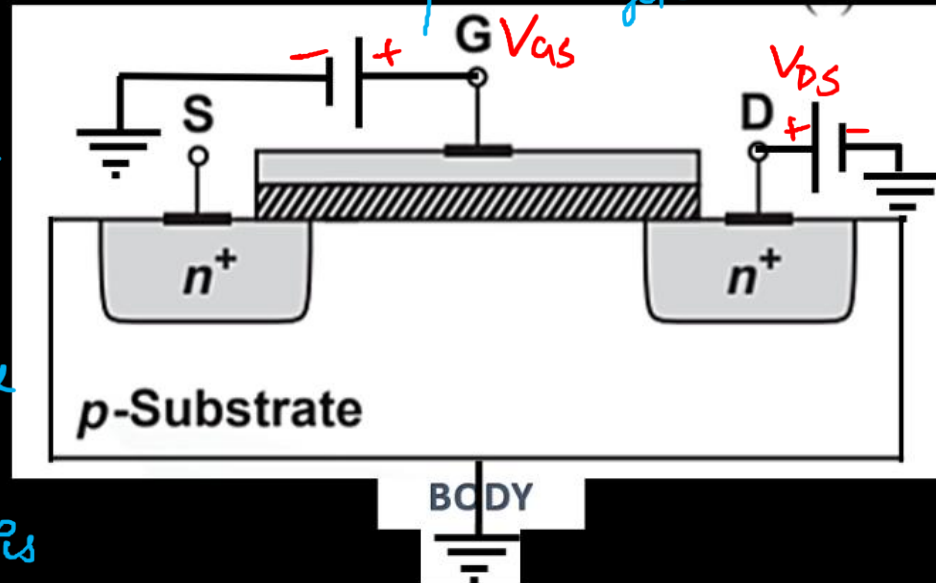
$$[0 < V_G < V_{Tn}]$$

The Majority charge carriers i.e. Holes get depleted/repelled from the Near region of the channel; Making it ready for Conduction whenever

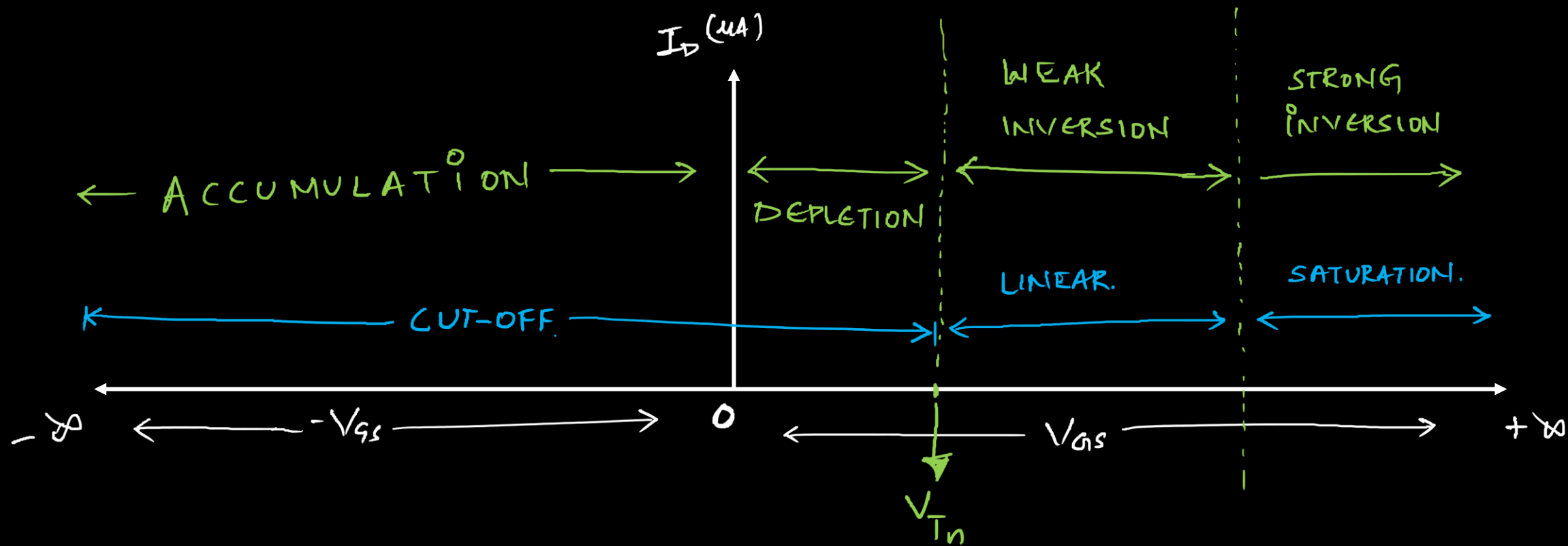
$V_{GS} > V_{Tn}$: LINEAR: Making an Easy

path for e^- Movement → CHANNEL FORMATION & CURRENT GENERATION

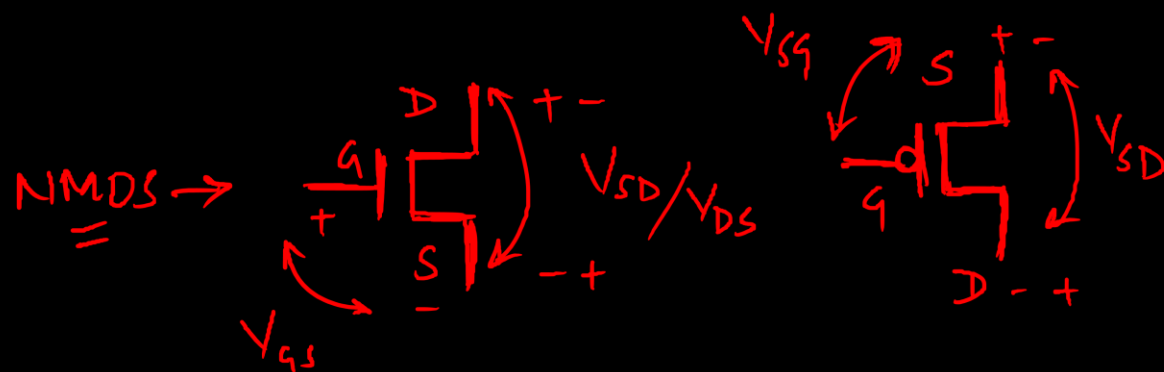
Sufficient Amount of V_G is Applied
@ G terminal $> V_{Tn}$ where the CHANNEL gets INVERTED ...

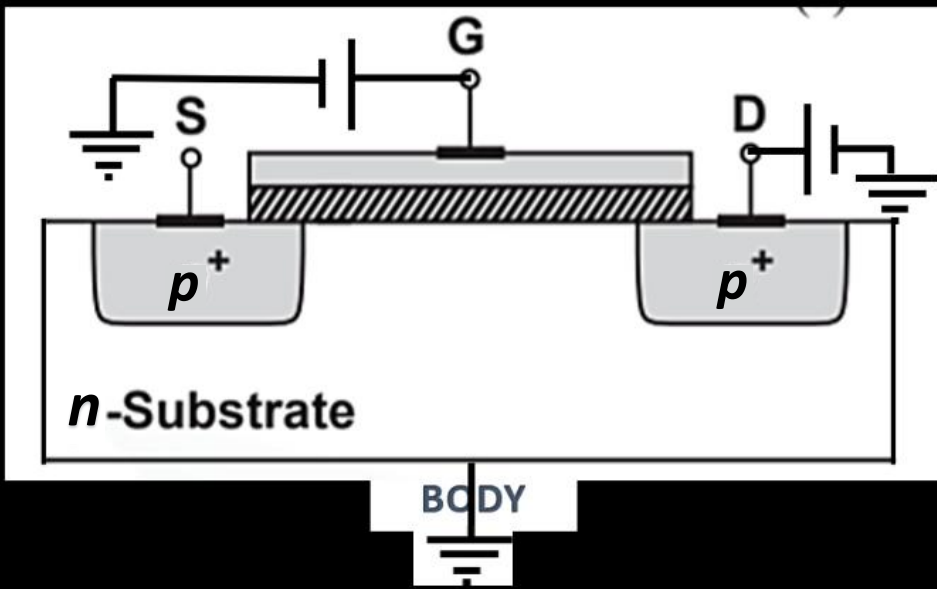


INVERSION



- i. CUT-OFF : $V_{GS} < V_{tn}$ ✓
- ii. LINEAR : $V_{GS} - V_{tn} > V_{DS}$ ✓
- iii. SATURATION : $V_{GS} - V_{tn} \leq V_{DS}$ ✓

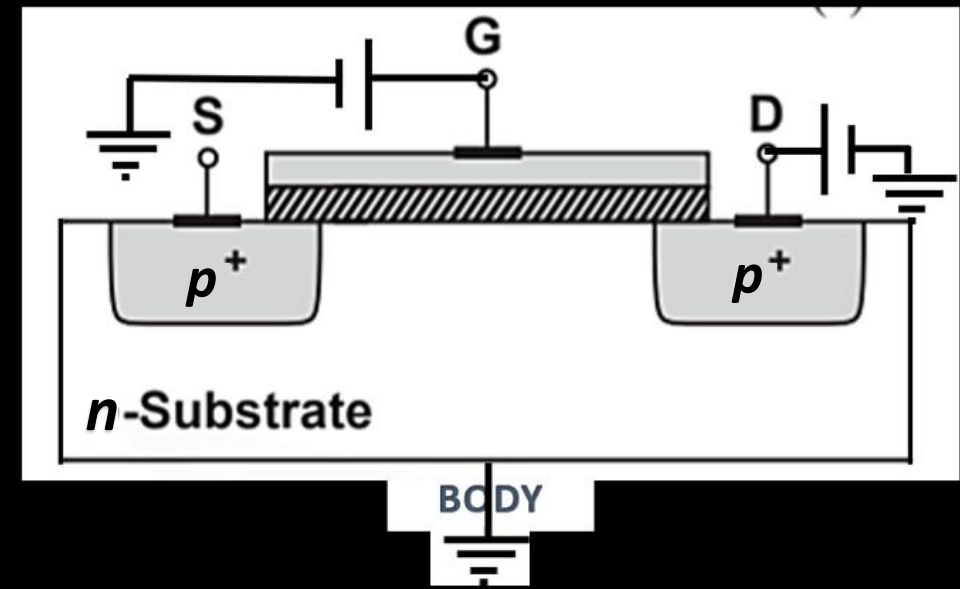




ACCUMULATION

$$V_{SG} +ve$$

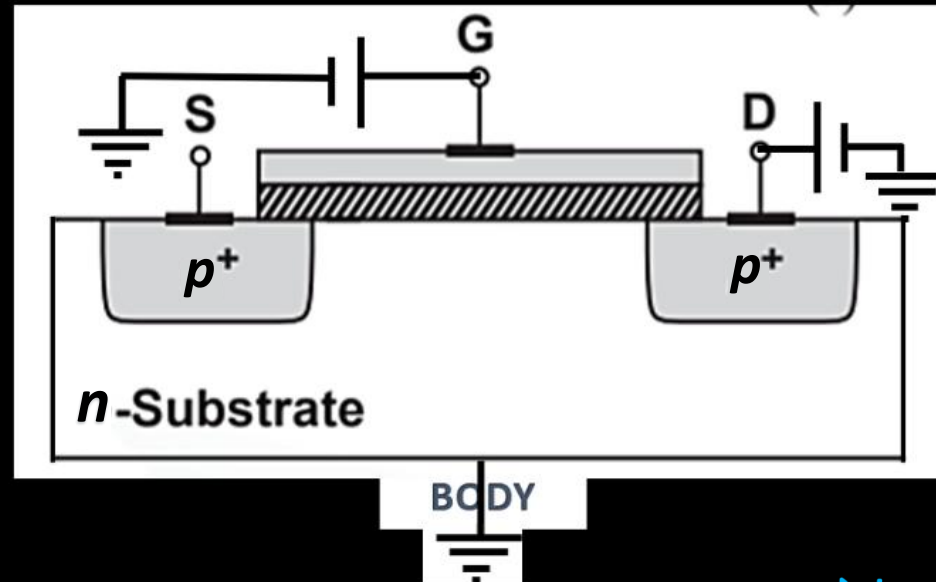
PMOS



DEPLETION

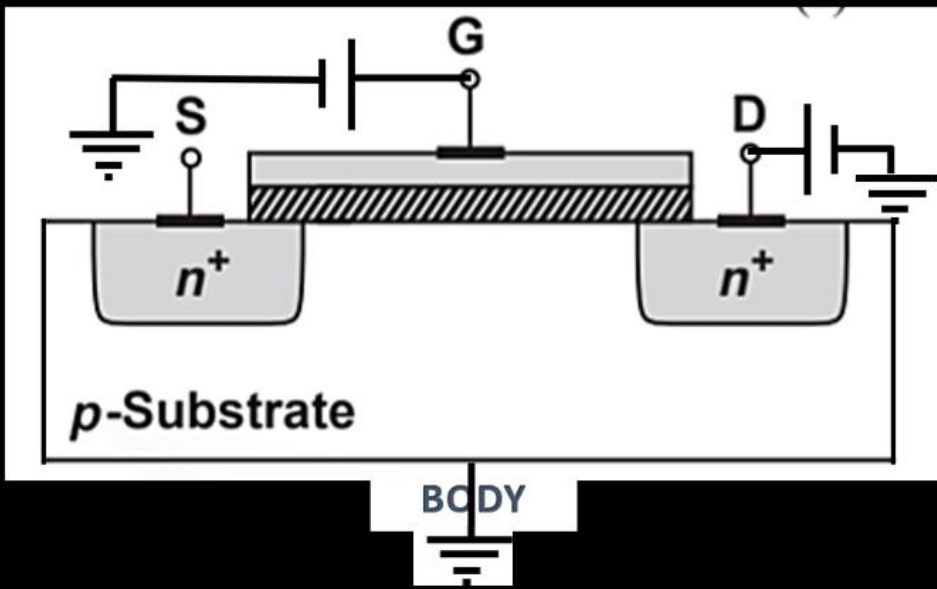
$$V_{tp} < V_{SG} < 0, ov$$

$$0 < V_{SG} < |V_{tp}|$$



INVERSION

$$V_{SG} > |V_{tp}|$$



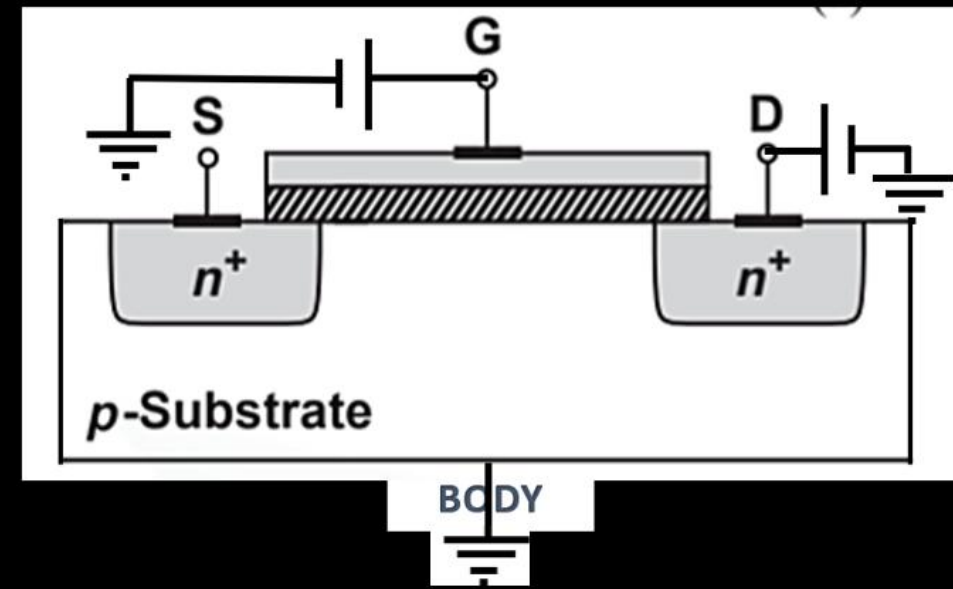
CUT OFF ✓

$$V_{GS} < V_{tn}$$

$$(I_D)_{lin} = \mu_n C_{ox} \frac{W}{L} \left\{ (V_{GS} - V_{tn}) \cdot V_{DS} - \frac{V_{DS}^2}{2} \right\}$$

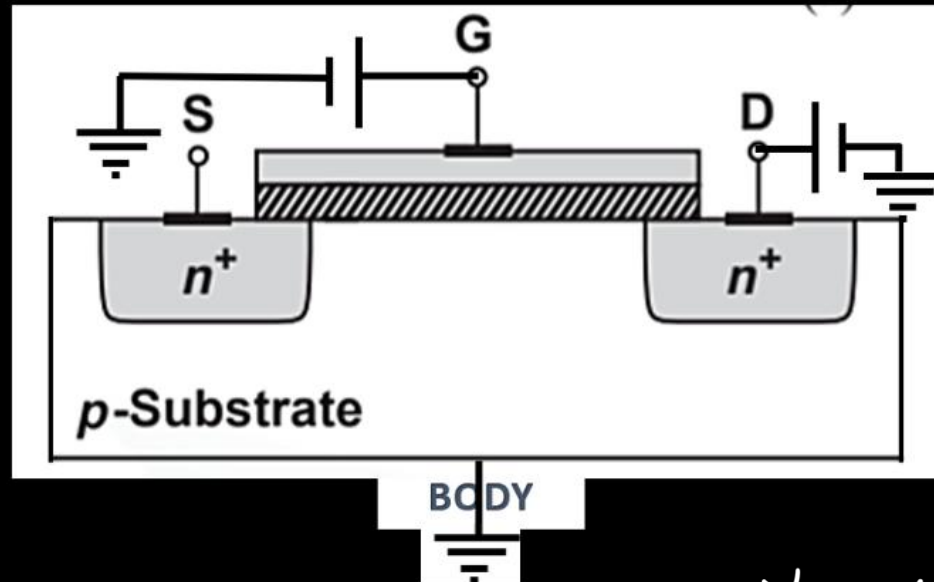
$$(I_D)_{sat} = \mu_n C_{ox} \frac{W}{2L} \left\{ V_{GS} - V_{tn} \right\}^2$$

NMOS



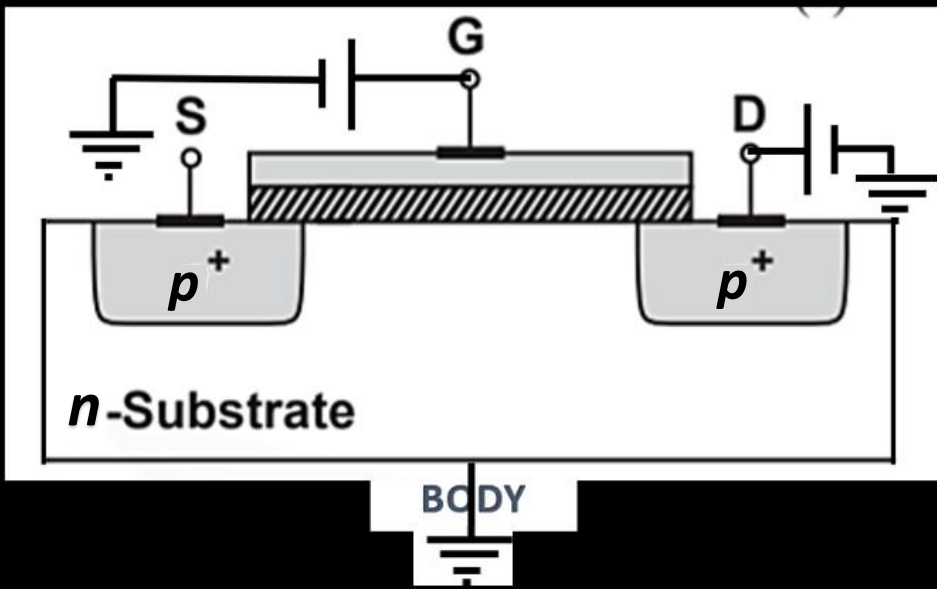
LINEAR

$$\checkmark V_{GS} > V_{tn}, \text{ or, } V_{GS} - V_{tn} > V_{DS} \checkmark$$



SATURATION

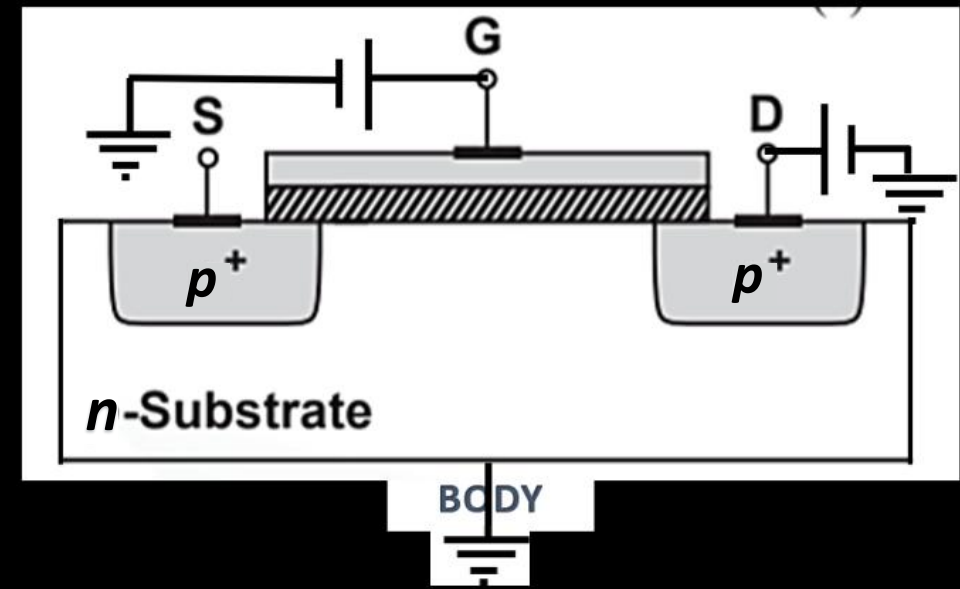
$$V_{GS} - V_{tn} \leq V_{DS}$$



CUT OFF

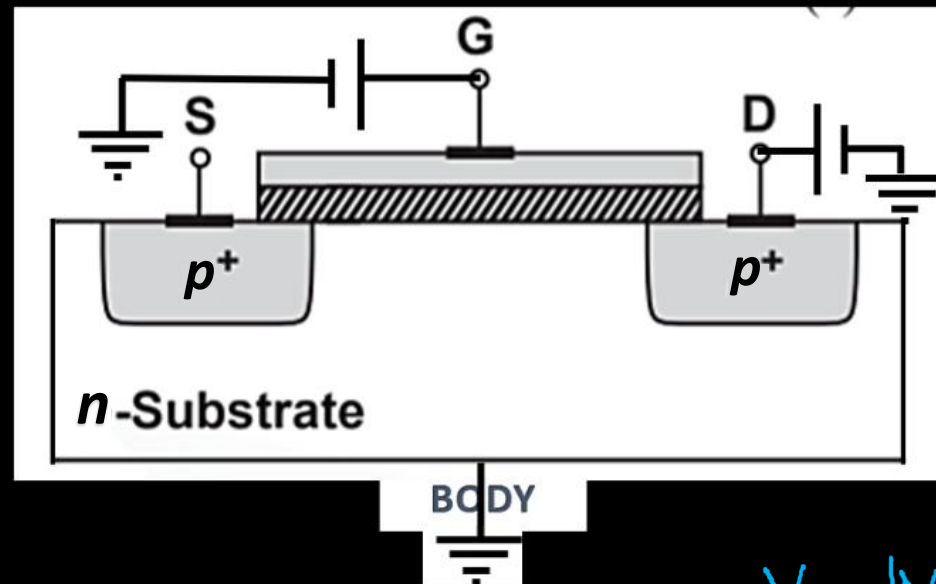
$$V_{SG} < |V_{tp}|$$

PMOS



LINEAR

$$V_{SG} > |V_{tp}| \quad \text{or} \\ V_{SG} - |V_{tp}| > V_{SD}$$

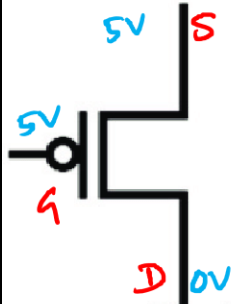
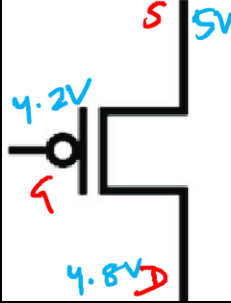
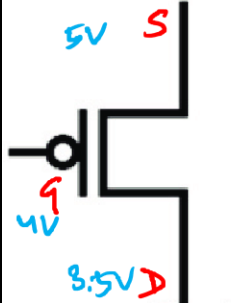
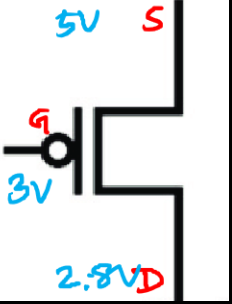
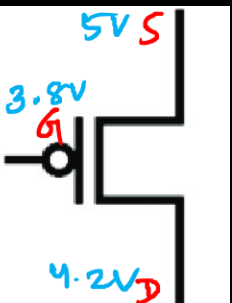
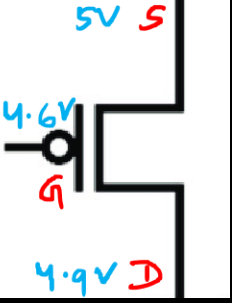
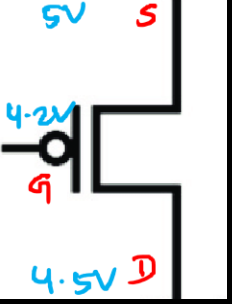
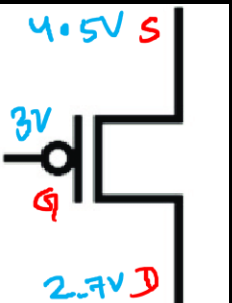
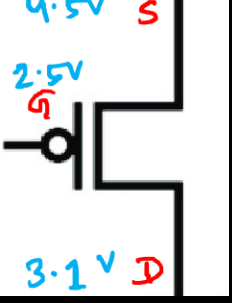


SATURATION

$$V_{SG} - |V_{tp}| \leq V_{SD}$$

I. What are the regions of operation of the following MOSFET configurations ?

Assume $V_{tn} = 0.4V$

- (a)  $V_{SG} = 5 - 5 = 0V$ ✓
 $V_{SD} = 5 - 0 = 5V$ ✓
 $|V_{tp}| = 0.4V$ ✓
 $V_{SG} - V_{tp} = 0V - 0.4V = -0.4V$ ✓
 $V_{SG} < V_{tp} = \boxed{\text{CUT-OFF}}$ ANSW.
- (b)  $V_{SG} = 5 - 4.2 = 0.8V$
 $V_{SD} = 5 - 4.8 = 0.2V$
 $V_{tp} = 0.4V$ **LINEAR**
 $V_{SG} - V_{tp} = 0.4V < V_{SD}$
- (c)  $V_{GS} = 5 - 4 = 1V$
 $V_{SD} = 1.5V$ ✓
 $V_{tp} = 0.4V$ **SATURATION**
 $V_{SG} - V_{tp} = 0.6V < V_{SD}$
- (d)  $V_{SG} = 5 - 3V = 2V$
 $V_{SD} = 5 - 2.8V = 2.2V$
 $V_{tp} = 0.4V$ **SATURATION**
 $V_{SG} - V_{tp} = 1.6V < V_{SD}$
- (e)  $V_{SG} = 5 - 3.8 = 1.2V$
 $V_{SD} = 0.8V$
 $V_{tp} = 0.4V$ **SATURATION**
 $V_{SG} - V_{tp} = 0.8 = V_{SD}$
- (f)  $V_{SG} = 0.4V$
 $V_{SD} = 0.1V$ **SATURATION**
 $V_{tp} = 0.4V$
 $V_{SG} - V_{tp} = 0V < V_{SD}$
- (d)  $V_{SG} = 5 - 4.2 = 0.8V$
 $V_{SD} = 0.5V$
 $V_{tp} = 0.4V$ **SATUR**
 $V_{SG} - V_{tp} = 0.4 < V_{SD}$
- (e)  $V_{SG} = 4.5 - 3V = 1.5V$
 $V_{SD} = 1.8V$
 $V_{tp} = 0.4V$ **SAT**
 $V_{SG} - V_{tp} = 1.1 < V_{SD}$
- (f)  $V_{SG} = 4.5 - 2.5 = 2V$
 $V_{SD} = 4.5 - 3.1 = 1.4V$
 $V_{SG} - V_{tp} = 2 - 0.4 = 1.6V > V_{SD}$
LINEAR.

II. What are the regions of operation of the following MOSFET configurations ?

Assume $V_{tn} = 0.4V$

